LED MEMORY GAME

Interactive Electronics Learning System

Group 19 · Project Report

Group Members

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1. Problem Statement

Traditional memory games are not sufficiently interactive for effectively teaching and reinforcing electronics concepts. Students often struggle to engage with abstract principles of embedded systems without hands-on, real-time feedback. The challenge was to design an interactive game using an Arduino microcontroller that would make learning electronics more engaging, practical, and enjoyable.

2. Methodology

The project was implemented using Arduino-based hardware and software. The team adopted a systematic approach, outlined below:

- **Hardware Setup:** Arduino Uno was used as the core microcontroller, paired with 4–6 LEDs, push buttons, resistors, a breadboard, connecting wires, and an LCD Display.
- **Pattern Generation:** LED sequences were programmed and stored within the Arduino code to produce randomised, escalating patterns for the player to memorise.
- **Player Interaction:** Players replicate the displayed LED sequences by pressing the corresponding push buttons, testing recall and reaction.
- **Difficulty Scaling:** The game automatically increases in difficulty after each successfully completed round, extending the LED sequence length.
- **Status Display:** The LCD screen provides real-time game status updates including instructions, scores, and feedback to the player.

3. Code & Schematic Overview

The project integrates both hardware schematics and Arduino firmware to achieve the desired functionality.

- **Code Structure:** The Arduino sketch manages LED output sequences, reads button inputs, tracks the player's progress, and communicates results to the LCD. Logic blocks handle sequence generation, input comparison, score tracking, and difficulty adjustment.
- **Schematic:** The schematic diagram illustrates the wiring of all LEDs, buttons, and the I2C LCD to the Arduino Uno. Resistors are included in series with each LED to limit current and protect components.
- **I2C LCD Integration:** An I2C LCD module was selected for compactness and ease of wiring, requiring only two data lines (SDA and SCL) rather than the multiple pins used by a parallel LCD.

4. Simulation & Demonstration

Simulation was conducted using **Tinkercad**, a browser-based electronics simulation platform. The results confirmed the following:

- The LED sequence functioned correctly, producing the intended visual patterns.
- Push buttons responded accurately and registered player inputs without error.
- The LCD displayed game instructions, status updates, and scores as designed.
- The difficulty progression behaved as expected, with sequences increasing in length after each successful round.

5. Deductions from the Simulation

Analysis of the simulation results led to the following key deductions:

- The Arduino platform is well-suited for building interactive, hardware-driven games.
- LED patterns and button inputs can be reliably synchronised within embedded C code.
- The I2C LCD effectively communicates game state to the user without excessive pin usage.
- Progressive difficulty is a viable mechanism for maintaining player engagement and promoting incremental learning.

6. Conclusion

Summary of Findings

- Arduino can be effectively used to build interactive, educational games.
- The LED memory game successfully demonstrates hardware and software integration in an embedded systems context.
- The project reinforces core electronics concepts including GPIO control, serial communication, and conditional logic.

Challenges Encountered

The primary challenge encountered was the difficulty in procuring required hardware components. The Adafruit-based LCD, originally planned for displaying high scores, was unavailable due to scarcity. The team adapted by switching to an I2C LCD module, which necessitated a complete overhaul of the initial codebase to accommodate the different communication protocol and library.

Proposed Improvements

- Implement non-volatile high score storage so scores persist after the microcontroller is powered off.
- Introduce additional difficulty levels to maintain long-term player engagement.

- Incorporate more varied and pleasant buzzer sounds for better audio-visual feedback.
- Improve internal wiring for cleaner, more professional circuit assembly.

7. References

1. Arduino LED Chaser — reference design for LED sequencing logic.
2. Simon Memory Game — conceptual inspiration for gameplay and difficulty mechanics.
3. Claude AI and ChatGPT — used to assist in refining and debugging the Arduino code.